

EDUCATION

The University of Toledo, Master of Science in Computer Science **GPA: 3.96** **Aug 2021 – Aug 2023**
Tribhuvan University, Bachelor in Computer Engineering **Sep 2013 – Oct 2017**

WORK EXPERIENCE

Software Developer - Blink Voice, Uniondale, NY **March 2024 - Present**

- Led development of an enterprise softphone (**VoIP/SIP**), shipped across Electron desktop, web, and Microsoft Teams clients..
- Built a real-time AI call assistant with live transcription and dynamic prompt injection over WebRTC data channels.
- Designed and implemented an **LLM** powered pipeline for automated call transcription, summarization, and topic extraction.
- Developed Express.js backend services for JWT/OAuth 2.0 auth, call analytics, and Redis pub/sub real-time event delivery.
- Built multi-tenant CRM integrations (HubSpot, Salesforce, Zendesk, Pipedrive) with OAuth 2.0 and encrypted API-key flows.
- Architected PostgreSQL schemas for analytics and AI usage metering; deployed microservices on GKE with Docker and CI/CD.
- Built a voice **AI IVR** on the OpenAI Realtime API and WebRTC, automating customer support triage and intelligent call routing.
- Developed a cross-platform social app (PitchMe) with Flutter, NestJS, and Firebase — feeds, follows, and real-time sync.
- Implemented live streaming and head-to-head battle rooms with real-time audience interaction via GetStream.io and Firebase.
- Integrated real-time AI sales coaching using ElevenLabs voice synthesis and the OpenAI API for simulated training calls.

UToledo College of Medicine and Life Sciences (COMLS), Toledo, OH

Software Engineer (Research Collaborator) **Oct 2023 – March 2024**

- Led **PREPARE** platform redesign and performance optimization using **Python (FastAPI)** to improve scalability and reliability.
- Scaled infrastructure with Docker and Kubernetes to support concurrent simulation sessions.
- Implemented data pipeline to process real-time patient simulation data, integrating **LLMs** for automated learner feedback .
- Integrated **open-source LLMs** for automated scenario generation and real-time event recognition in medical training simulations.

Research Assistant (Software Developer) **Aug 2021 – Aug 2023**

- Collaborated with cardio-thoracic researchers on **ML** models for heart surgery data analysis and clinical outcome prediction.
- Built an **NLP-Enabled** multimodal assessment platform **PREPARE** for simulation-based medical education.
- Implemented backend microservices (**Python, Node.js**) with REST APIs supporting multimodal data processing and integration.
- Developed **React.js** web client for curriculum management, model training workflows, and learner performance dashboards.
- Integrated multiple speech-to-text libraries, developed services for model training, event detection, and audio transcription.

Software Engineer Intern - Quantiphi Inc, Marlborough, MA **May 2022 - August 2022**

- Contributed to development of **Fargo**, a Virtual Agent for **Wells Fargo**, enhancing the banking experience for 70+ million customers.
- Collaborated with global **SCRUM** teams and multi-team stakeholders (linguists, developers, product) to design and launch features.
- Supported code reviews, validation, and testing, including unit testing and debugging, to improve system reliability.

Software Engineer – FOCUSONE Nepal Pvt.Ltd, Kathmandu, Nepal **July 2018 – May 2020**

- Developed Vehicle Tax Calculator (VTAX), delivering backend services and a responsive front-end for vehicle tax processing.
- Built **RESTful** APIs using **PHP** and designed efficient **MySQL** databases to support reliable backend operations.
- Created responsive user interfaces with **React.js, HTML**, and **CSS** to improve usability.

PUBLICATION AND PRESENTATION

- **Automated Multimodal Performance Evaluation in Simulation-based Medical Education using Natural Language Processing** published and presented a poster at **ACM/IEEE 14th International conference on Cyber-Physical Systems**, May 9-12 | SA, Texas, 2023
- **Improved Multimodal Data Acquisition and Synchronization through NLP Enabled Event Detection in SBME | MS THESIS | Aug 2023**
- Co-authored a scientific abstract **Role of Smart Wearable Devices in Postoperative Care of Cardiac Surgery Patients**, presented at the **59th Annual Meeting of The Society of Thoracic Surgeons**, Jan 21-23| San Diego, California, 2023

TECHNICAL SKILLS

Programming/Scripting:	JavaScript (JS), Python, PHP, C, C++
Frontend/Mobile:	React.Js, Flutter, Electron, HTML, CSS
Backend/Database:	Node.js, Express.js, NestJS, FAST API, PostgreSQL, MySQL, Firebase
Cloud/DevOps:	AWS, GCP, GKE, Docker, Kubernetes, CI/CD Pipelines
Data Science & Machine Learning:	LLMs, TensorFlow, Keras, PyTorch, Transformer, BERT, LSTM, CNN, GAN